CERTPATH PUBLISHING · FORMULA CHEAT SHEET

CAST Exam — Quick Reference

EEI Construction and Skilled Trades Test · certpathpublishing.store

⚡ ELECTRICAL & POWER

Ohm's Law & Power

$$V = I \times R$$

$$P = V \times I = I^2R = V^2/R$$

- **Series:** $R = R_1 + R_2 + R_3$ (same current)
- **Parallel:** $1/R = 1/R_1 + 1/R_2$ (same voltage)
- Ohm = volts dropped per amp of current.

Work, Force, Energy, Power

$$W = F \times d$$

$$P = W / t$$

$$KE = \tfrac{1}{2}mv^2 \quad \cdot \quad PE = mgh$$

- 1 hp = 550 ft·lb/s = 746 W
- 1 ft·lb ≈ 1.356 J

🔧 MECHANICAL ADVANTAGE

Simple Machines — MA

Machine	MA Formula
Lever	L_{effort} / L_{load}
Pulley (block & tackle)	# supporting ropes
Inclined plane	Length / Height
Wheel & axle	R / r
Screw	2πr / pitch

Trade force for distance · never both.

Friction

$$f = \mu \times N$$

- **Static > Kinetic** — harder to start than to keep moving.
- Rolling friction is much lower than sliding.
- μ depends on surfaces, not area or weight.

📐 MATH & UNITS

Area & Volume

- **Rectangle:** A = L × W
- **Triangle:** A = ½bh
- **Circle:** $A = \pi r^2$; $C = 2\pi r$
- **Cylinder V:** $\pi r^2 h$ · **Cone V:** $\frac{1}{3}\pi r^2 h$
- **Box V:** L × W × H

Unit Conversions

- 1 ft = 12 in = 30.48 cm
- 1 gal = 231 in³ ≈ 3.785 L
- 1 kW = 1.341 hp · 1 hp = 746 W
- 1 ft·lb ≈ 1.356 J
- °F = (°C × 9/5) + 32
- °C = (°F – 32) × 5/9

📖 READING COMPREHENSION

Skim First

- Read the question, **then** scan for keywords.
- Focus on what's asked — skip filler.

Watch for Signals

- **However** → contrast coming.
- **Therefore** → conclusion.
- **Except** → pick the odd one out.

Eliminate Extremes

- "Always" / "Never" answers are often wrong.
- Pick the answer the passage **directly** supports.

- Don't memorize; verify.
- **Primarily** → main idea, not detail.
- Outside knowledge is a trap.

🎯 **Test Day Plan:** CAST is ~87 min total. Target ≤**45 sec** per question on the first pass — flag and return. **Bring:** photo ID, confirmation, water. **Leave behind:** phone, smart watch, notes, calculators not listed as approved.